

No. My SQL Logical Questions

Salary:- Maximum salary

Salary:- Nth Highest salary

Salary:- Top Nth salary

salary + department:- Department-wise Total Salary

salary + department:- Highest Salary ka Department kaun sa hai?

salary + department:-Department Having Total Salary > 100000

salary + department:- Department-wise average salary?

Duplicate values in a Table?

Find Maximum Salary

emp_name	salary
AMIT	50000
ROHIT	90000
NEHA	90000
PRIYA	65000

- Sirf salary chahiye

```
SELECT MAX(salary) AS highest_salary FROM employees;
```

max_salary
90000

- Employee with Maximum Salary (Name + Salary chahiye)

```
SELECT emp_name, salary FROM employees
WHERE salary = (
    SELECT MAX(salary) FROM employees
);
```

emp_name	salary
ROHIT	90000
NEHA	90000

Find 3rd Highest Salary

emp_name	salary
AMIT	50000
ROHIT	90000
NEHA	70000
PRIYA	80000
RAVI	60000

Using Limit

```

SELECT salary FROM employees ORDER BY salary DESC LIMIT 2,1;
SELECT emp_name, salary FROM employees ORDER BY salary DESC LIMIT 2,1;

-- (OR) Standard style
SELECT salary FROM employees ORDER BY salary DESC LIMIT 1 OFFSET 2;

-- For DISTINCT duplicate salary ko handle karega
SELECT DISTINCT salary FROM employees ORDER BY salary DESC LIMIT 1 OFFSET 2;

```

Using Subquery

```

SELECT MAX(salary) AS third_highest_salary
FROM employees
WHERE salary < (
    SELECT MAX(salary)
    FROM employees
    WHERE salary < (
        SELECT MAX(salary)
        FROM employees
    )
);

```

third_highest_salary
70000

Employee name ke sath 3rd Highest Salary

- Using Sub Query

```
SELECT emp_name, salary
FROM employees
WHERE salary = (
    SELECT MAX(salary)
    FROM employees
    WHERE salary < (
        SELECT MAX(salary)
        FROM employees
        WHERE salary < (
            SELECT MAX(salary)
            FROM employees
        )
    )
);
```

emp_name	salary
NEHA	70000

- Agar Duplicate hai to dono print honge

emp_name	salary
NEHA	70000
RAVI	70000

Using DENSE_RANK() (Most Preferred for Experienced Interview)

Using COUNT() Subquery

Find Top n Salaries

Find Top 4 Salaries

- Using Limit

```
SELECT DISTINCT salary FROM employees ORDER BY salary DESC LIMIT 4;

-- with name isme distinct nahi aayega
SELECT emp_name, salary FROM employees ORDER BY salary DESC LIMIT 4;
```

- DISTINCT salaries ke saath employees ke names bhi chahiye to using Subquery

```
SELECT emp_name, salary
FROM employees
WHERE salary IN (
    SELECT DISTINCT salary
    FROM employees
    ORDER BY salary DESC
    LIMIT 4
)
ORDER BY salary DESC;
```

- Using Subquery

```
SELECT emp_name, salary
FROM (
    SELECT emp_name,
           salary,
           DENSE_RANK() OVER(ORDER BY salary DESC) AS rnk
    FROM employees
) AS temp
WHERE rnk <= 4;

-- In Oracle
SELECT SAL FROM(SELECT DISTINCT SAL FROM EMP WHERE SAL IS NOT NULL ORDER BY SAL
DESC)WHERE ROWNUM <6;
```

Department salary Question

emp_name	department	salary
AMIT	IT	50000
ROHIT	HR	60000
NEHA	IT	70000
PRIYA	HR	80000
RAVI	SALES	40000

Department-wise Total Salary

```
SELECT department, SUM(salary) AS total_salary
FROM employees GROUP BY department;
```

department	total_salary
HR	140000
IT	120000
SALES	40000

Highest Salary ka Department kaun sa hai?

```
SELECT department, SUM(salary) AS total_salary FROM employees
GROUP BY department
ORDER BY total_salary DESC
LIMIT 1;
```

department	total_salary
HR	140000

Department Having Total Salary > 100000

```
SELECT department, SUM(salary) AS total_salary
FROM employees GROUP BY department
HAVING SUM(salary) > 100000;
```

department	total_salary
HR	140000
IT	120000

Department-wise average salary?

```
SELECT department, AVG(salary) AS avg_salary
FROM employees GROUP BY department;
```

department	avg_salary
HR	70000
IT	60000
SALES	45000

How to Find Duplicate values in a Table?

id	emp_name	email
1	Mohit	mohit@gmail.com
2	Rahul	rahul@gmail.com
3	Amit	mohit@gmail.com
4	Sumit	sumit@gmail.com
5	Raj	rahul@gmail.com

```
-- COUNT(*) Null value ko bhi count karta hai
SELECT email, COUNT(*) AS count FROM employees GROUP BY email
HAVING COUNT(*) > 1;

-- COUNT(Email) NULL valu ko count nahi karta hai
SELECT Email, COUNT(Email) AS total_email FROM employees GROUP BY Email
HAVING COUNT(Email) > 1;
```

Email	total_email
mohit@gmail.com	2
rahul@gmail.com	2

```
-- NULL values mtb
```

email
mohit@gmail.com
rahul@gmail.com
NULL
sumit@gmail.com
rahul@gmail.com

```
-- If you want the complete records of duplicates
SELECT *
FROM employees
WHERE email IN (
    SELECT email
    FROM employees
    GROUP BY email
    HAVING COUNT(*) > 1
);
```


